

Name	Index Number	CTG

YISHUN JUNIOR COLLEGE JC 2 SEMESTRAL EXAMINATION 2018

BIOLOGY

8876/01

HIGHER 1

14th SEPTEMBER 2018
Fri 0800 – 0900

1 hour

Paper 1 Multiple Choice

Additional material: Optical Mark Sheet

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READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your NRIC number, name and CTG on the Optical Mark Sheet in the spaces provided.

There are **twenty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.

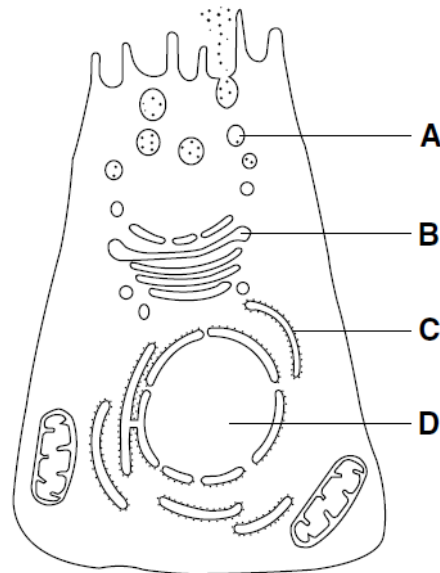
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This question paper consists of 9 printed pages and 1 blank page.

- 1 The diagram is taken from an electron micrograph of a cell which secretes digestive enzymes.

Where are these enzymes made? **Ans: C**



Explanation: digestive enzymes are proteins. Proteins are translated by 80S ribosomes bounded to ER (RER).

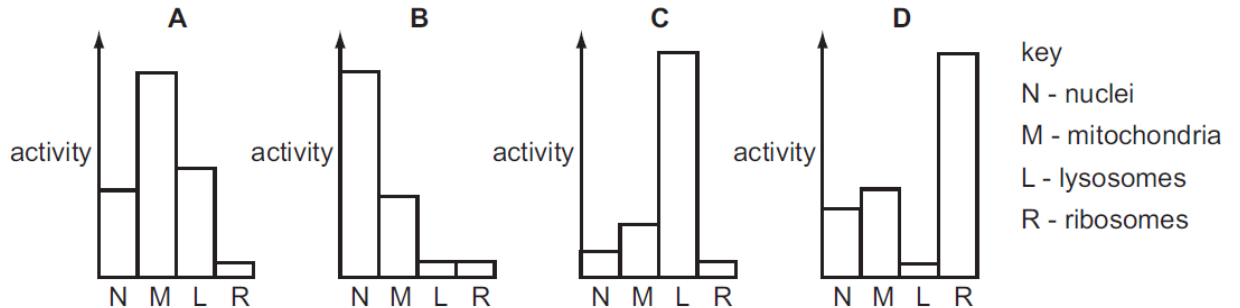
- 2 What is the order of size of cell components?

	Largest → Smallest			
A	Mitochondria	Ribosomes	Starch grains	Nuclei
B	<u>Nuclei</u>	<u>Chloroplasts</u>	<u>Mitochondria</u>	<u>Ribosomes</u>
C	Ribosomes	Mitochondria	Chloroplasts	Starch grains
D	Starch grains	Mitochondria	Chloroplasts	Ribosomes

Explanation: Nucleus has the largest molecular weight while ribosomes have the smallest molecular weight.

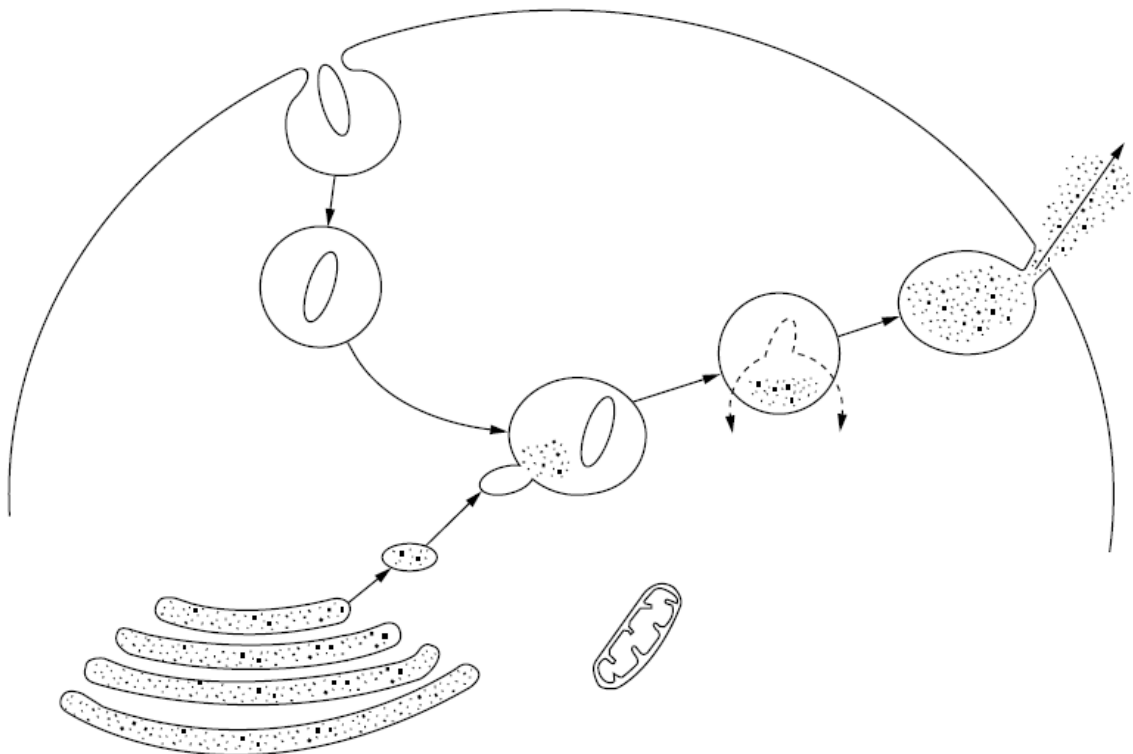
- 3 A piece of mammalian tissue was homogenised and subjected to differential centrifugation to yield four subcellular fractions. The activity within each fraction, of four different types of enzyme, **A**, **B**, **C** and **D**, was investigated.

Which bar chart shows the results of investigating hydrolytic enzyme activity?



Explanation: hydrolytic enzymes are found in the lysosomes. Therefore, the bar with the highest activity for lysosomes should reflect the investigative outcome for hydrolytic enzymes.

- 4 The diagram shows several processes taking place in a cell.



Which processes are shown in the diagram and involve the cell surface membrane of the cell?

- A** active transport and diffusion
B diffusion and osmosis

C endocytosis and exocytosis

D endocytosis and osmosis

Explanation: Endocytosis is in the intake of materials into the cell via the formation of the pseudopods and engulfment of materials in an endosome. Exocytosis is the discharge of materials via the fusion of endosome with the cell surface membrane.

- 5** A plant cell is placed in a solution with a less negative (higher) water potential than the cell contents.

Which change occurs in the cell and what causes the change?

	change	cause
A	cell becomes more flaccid	solution diffuses out of the cell
B	cell becomes more flaccid	water diffuses out of the cell
C	cell becomes more turgid	solution diffuses into cell
<u>D</u>	<u>cell becomes more turgid</u>	<u>water diffuses into cell</u>

Explanation: If solution water potential is higher than cell content, water will diffuse into the cell via osmosis causing the cells to become turgid.

- 6** Beetroot cells contain a water-soluble red pigment. Two test tubes were set up as described in the table.

tube 1	Pieces of washed raw beetroot in water
tube 2	Pieces of washed raw beetroot in water containing 3 drops of cyanide, a respiratory inhibitor.

After 30 minutes, the water in tube 2 contained a red pigment but the water in tube 1 did not.

Which of the following statements are false for tube 2?

- 1 Pigment molecules passed out and were replaced by cyanide.
- 2 The cell membrane was unable to retain the red pigment.
- 3 Water entered the tissue by osmosis and caused the cells to burst.
- 4 Water passed out of the cells by osmosis and carried the soluble pigment with it.
- 5 The same result will occur if ethanol was used instead of cyanide.

- A 1 and 3 only
- B 3 and 4 only
- C 2 and 5 only
- D 1, 3 and 4 only

Explanation: deduce what is true – the reason why tube 2 is red is because the red pigments leak out of the cell due to the disruption of the cell membrane (#1 and #5). Beetroot cells are plant cells; plant cells have cell wall; therefore, osmosis should not cause the cells to burst (contradict #3). Cyanide disrupts the membrane and do not replace the red pigments (contradict #1). Red pigments are too big to diffuse across the membrane (contradict #4).

7 Some features of haemoglobin are listed.

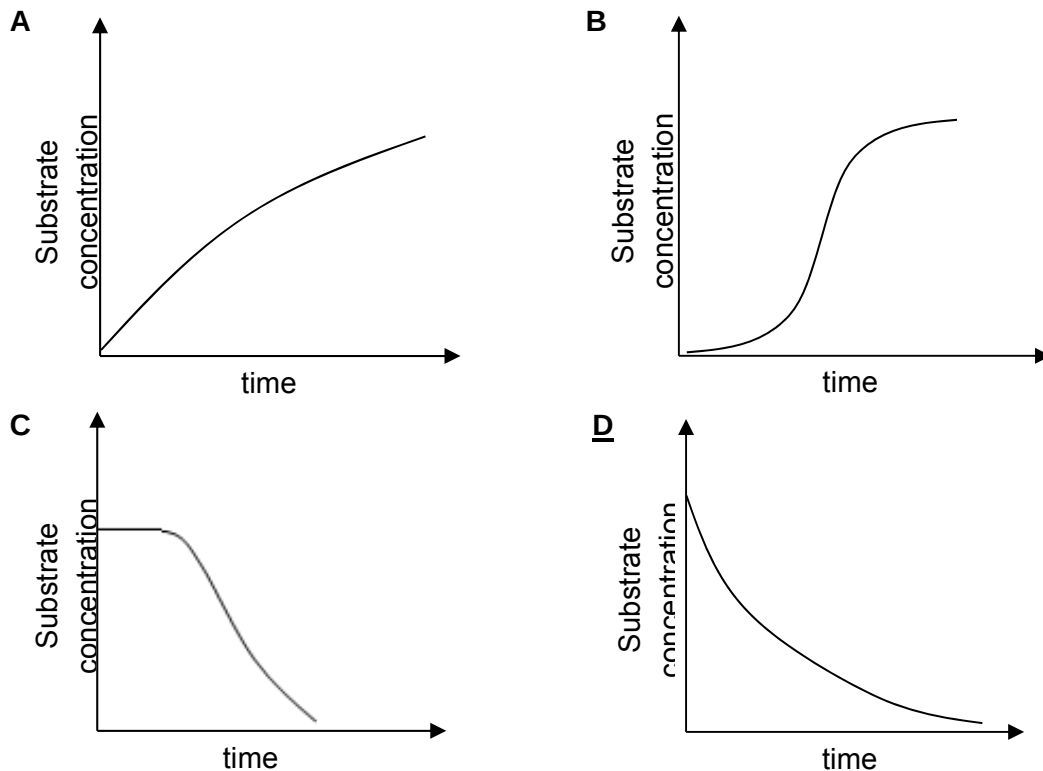
- 1 They are made up of a protein component called globin and a non-protein component called haem group.
- 2 There are two types of polypeptide chains: the α -globin chain and the β -globin chain.
- 3 Each polypeptide chain is held by hydrophobic interactions, hydrogen and ionic bonds and disulfide bonds.
- 4 Most of its hydrophilic polar amino acid residues are on the external surface of the globular structure.

Which are these statements are true ?

- A 1 and 3
- B 1 and 2
- C 1, 2 and 4
- D 1 and 4

Comments : there are no disulphide bonds and polypeptide is α -globin chain

- 8 Which graph represents the changes in substrate concentration during the course of an enzyme catalysed reaction?



- 9 Which is a correct statement about obtaining human embryonic stem cells for research?
- A Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.
 - B The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.**
 - C The cells must be removed within a day following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.
 - D The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.

Explanation: A is not correct because it is not ethically accepted to remove ESCs from blastocyst as the blastocyst is deemed as a potential life. C is not correct because the cells removed within a day following successful fertilisation are zygotic stem cells. D is not correct because it is the trophoblast that will develop into a placenta, not the inner cell mass.

- 10** Mesenchymal stem cells can differentiate into several types of cells belonging to our skeletal tissue, such as cartilage, bone and fat.

Which statement correctly describes mesenchymal stem cells?

- A** They are specialised cells that can give rise to a variety of cell types.
- B** They can be stimulated by chemical signals to express certain genes.
- C** They lose genetic information as they differentiate.
- D** They occur in large numbers in the bone marrow.

Explanation: A is false because stem cells are not specialised cells. C is false because genetic information is condensed, not lost. D is false because haematopoietic stem cells are found in bone marrow.

- 11** Which statements about ribosomal RNA(rRNA) is true?

- 1 rRNA is involved in the binding of mRNA and not tRNA.
- 2 rRNA with associated proteins, fold into complex tertiary structures within the ribosome.
- 3 The 3' terminus of rRNA in prokaryotes is known to interact with the initiation region of mRNA via the Shine–Dalgarno sequence.
- 4 Nucleolus is the site of rRNA transcription and processing, and of ribosome assembly.

- A** 1 and 3 **B** 1, 2 and 3 **C** 2 and 4 **D** 2, 3 and 4

Explanation: 1 is false because tRNA anticodon is complementary to the codon of the mRNA, not rRNA. 2 is true because rRNA is a component of ribosome. 3 is true because the rRNA in the small ribosomal subunit binds to the initiation region of the mRNA. 4 is true because rRNA is transcribed in nucleolus.

12 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

- 1 The anti-parallel arrangement of the DNA strands.
- 2 The RNA primers are required to initiate DNA elongation.
- 3 DNA polymerase can join new nucleotides to the 3' end of the growing strand.
- 4 Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.

A 2 and

B 1 and 3

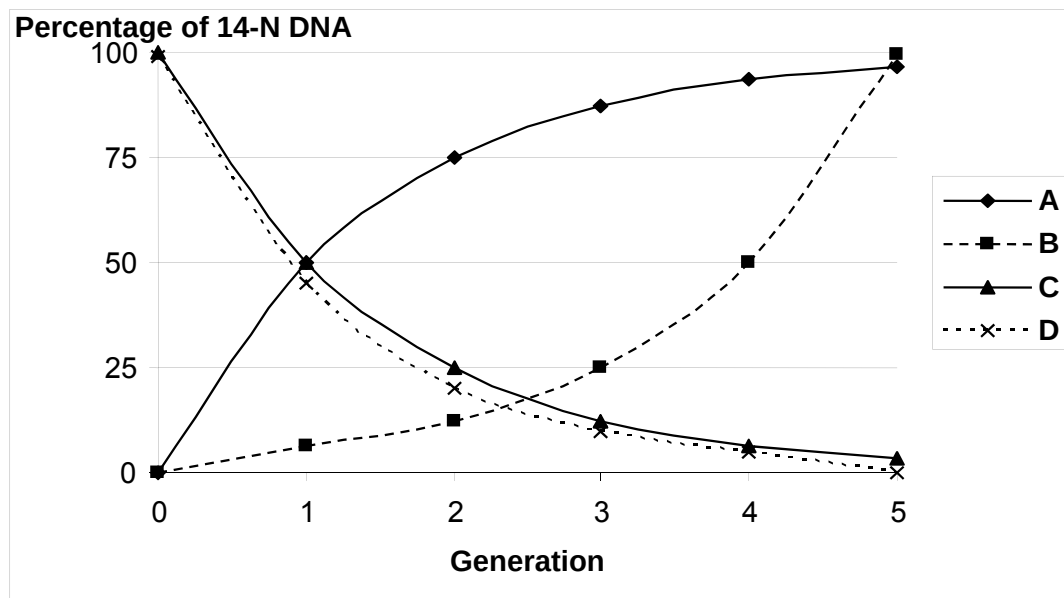
C 1 and 4

D 2 and 3

Explanation: 2 is false because both leading and lagging strands require primers to initiate DNA elongation. 4 is false because helicase and ssbp work on the 3' end of the DNA strand because daughter strand is elongated from 5' → 3'

13 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for 5 generations. The percentage of ^{14}N DNA strands in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative?



Answer A

^{14}N DNA increases. Between A or B . first generation – 50 % (2 molecules of $^{14}\text{N} : ^{15}\text{N}$)
Second generation : 2 molecules of $^{14}\text{N}^{15}\text{N}$ and 2 molecules of $^{14}\text{N}^{14}\text{N}$) $6/8 = 75\%$ etc

- 14 A new form of life is discovered. It has a genetic code much like that of organisms on Earth except that there are five different DNA bases instead of four and the base sequences are translated as doublets instead of triplets. How many different amino acids could be accommodated by this genetic code?
- A 10
- B 25**
- C 32
- D 64

Explanation: $5^2 = 25$

- 15 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both cause changes in number of chromosomes.	Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.
C	<u>Both affect DNA sequence.</u>	<u>Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.</u>
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

Option A : Gene mutations do not always produce dominant alleles. Option B : Gene mutation does not cause changes to the total number of chromosomes in an organism.

- 16 Which environmental factor is associated with the development of melanomas (a type of skin cancer)?
- A alcohol
- B tar
- C ultraviolet light
- D X-radiation

Explanation: melanomas are skin cancer, therefore, only UV aggravate the risk of cancer.

- 17 The table shows the DNA triplet codes for some amino acids from the strand complementary to mRNA.

Amino acid	DNA triplet code
Glycine	CCA, CCG, CCT, CCC
Leucine	AAT, AAC, GAA, GAG, GAT, GAC
Lysine	TTT, TTC
Methionine	TAC
Proline	GGA, GGG, GGT, GGC
Threonine	TGA, TGG, TGT, TGC

The sequence of DNA triplets from the strand complementary to mRNA form part of a gene is shown.

...TAC TTT AAT GGC CCT GAG GGC TAC TGT ...

Which mutated DNA sequences of this part of a gene would result in the same amino acid sequences as the original gene sequences ?

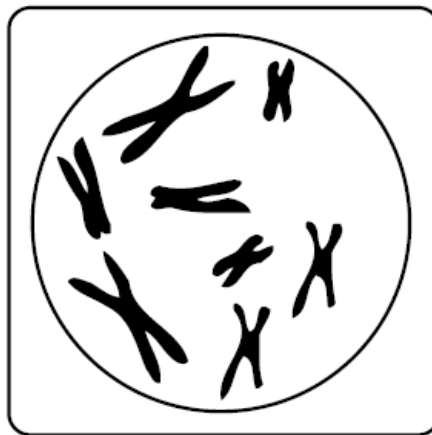
- A ...TAC TTT AAT GGC CCT GAG GGT CCA TGT
- B ...TAC TTC GAT GGC CCT GAG GGC TAC TGT
- C ...TAC TTT AAT GGC CCG GAG TGA TAC TGT
- D TAC TTT AAT GGC CCT GAG GGC TTC TGT

Explanation: no change in amino acid sequence → implies silent mutation

- 18** Cancer research has found that gene amplification is involved in the development of cancer. This research has identified one type of sequence that causes a break in DNA before gene amplification.

How might the findings of this research be used?

- A** to identify people at risk of developing cancer
 - B** to prevent cancer from developing in 'at-risk' people
 - C** to repair DNA breaks from amplification occurs
 - D** to use restriction endonuclease to remove DNA
- 19** The diagram shows a diploid cell during mitosis.

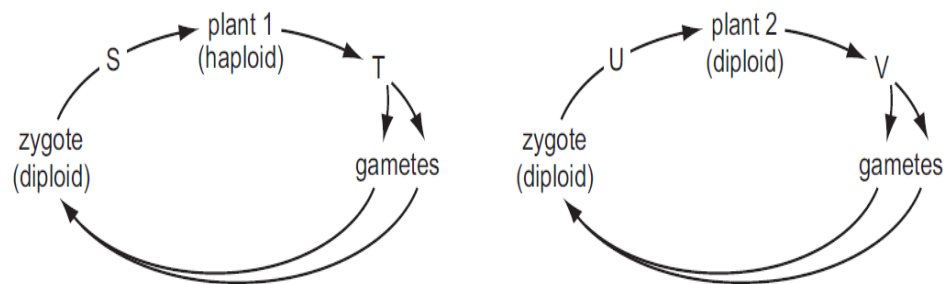


Which stage of mitosis is shown?

- A** anaphase
- B** metaphase
- C** prophase
- D** telophase

Explanation: Recall the behaviour of the chromosomes in the different stages of mitosis.

- 20 The diagram shows the life-cycles of two types of simple plant.



Where will reduction divisions occur in the life cycles?

- A at S and U
B at S and V
 C at T and U
 D at T and V

Explanation: reductive divisions occur when there is a need reduced number of chromosomes from diploid to haploid.

- 21 In corn, the colour of the stem can be either wild type, which is dark green, or virescent, which is pale green.

Cross	Phenotype of parents	Temperature at which raised	Offspring
1	virescent X virescent	20°C	virescent
2	virescent X virescent	37°C	wild type
3	wild type X wild type	20°C	3 wild type: 1virescent
4	wild type X wild type	37°C	wild type

The parents of crosses 1 and 2 had the same genotypes. The parents of crosses 3 and 4 had the same genotypes.

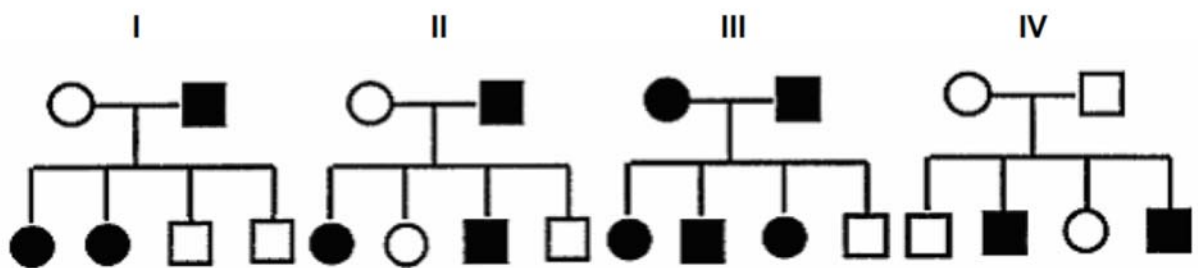
Which of the following can be deduced from the data?

- 1 Wild type is dominant.
 - 2 The wild type parents are heterozygotes.
 - 3 Virescent is the recessive phenotype but the characteristic will only be expressed at lower temperature.
 - 4 The parents of cross 1 are only recessive at 20°C.
- A 2 only
 B 1 and 3 only
C 1, 2 and 3 only

D all of the above

Comments cross 3 has offspring virescent at lower temperature and not at 37°C even though genotypes are the same. Statement 4 is incorrect as dominant or recessive not affected by temperature but rather the expression is affected

- 22 The figure below shows 4 pedigrees. A shaded circle represents an affected female while a shaded square represents an affected male.



Which pedigree(s) show(s) a sex-linked dominant trait in humans?

- A I only
B I and III
 C II and III
 D III and IV

Sex linked dominant :

I : shaded square male $X^D Y \rightarrow$ pass one X^D to female and diseased . Male not diseased

If diseased is recessive, $X^d Y \times X^D X^D$ female, then females should not be affected.

II : $X^D Y$ or $X^d Y$ shaded male. If dominant $X^D Y \rightarrow$ female diseased .

But male affected. Can be because of recessive allele from mother $X^d X^d$

Option III ;

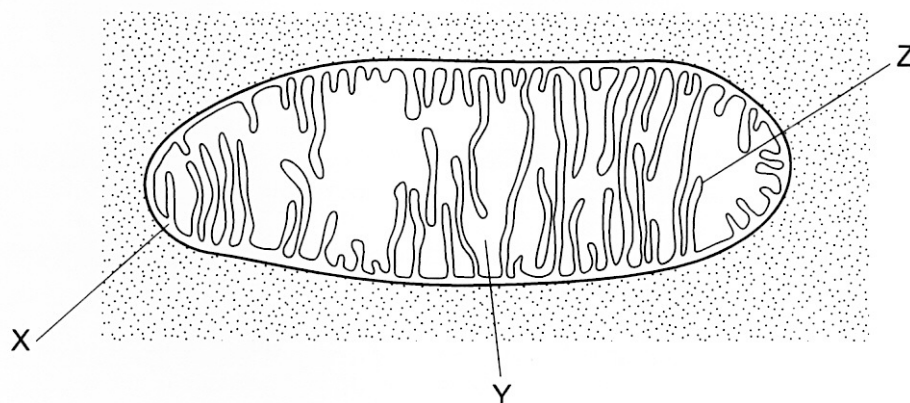
$X^d X^d \times X^d Y \rightarrow$ all males affected , But see one male unaffected

$X^D X^D \times X^D Y \rightarrow$ all sexes affected

$X^D X^d \times X^D Y \rightarrow$ females affected , males affected and males unaffected

Option IV : If dominant, at least one parents should be shaded

- 23 The electron micrograph shows an organelle found in a plant cell.

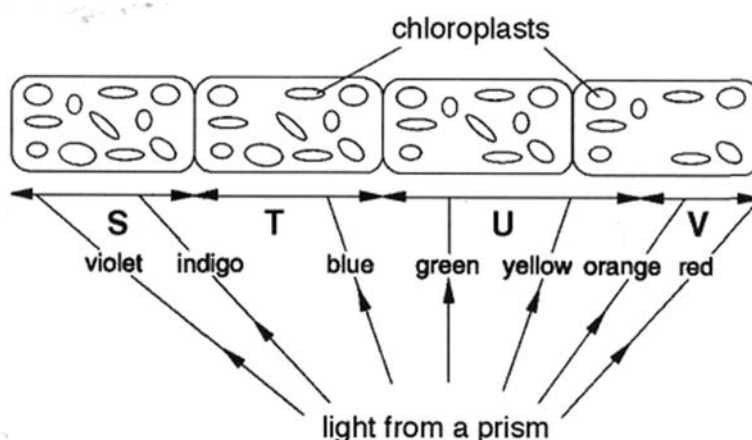


Which row shows the correct location of each named biological molecule?

	DNA	oxaloacetate	ATP
A	absent	Y	Z
B	Y	Z	Y
C	Y	Y	X
<u>D</u>	<u>Y</u>	<u>Y</u>	<u>Y</u>

Explanation: DNA is found as circular DNA in the matrix. Oxaloacetate is a substrate that is part of the Krebs cycle. ATP is produced during chemiosmosis into matrix.

- 24 A filament of the alga *Cladophora* was illuminated as shown in the diagram. Motile, aerobic bacteria were placed in the water with the alga and their position was determined by microscopic examination after 10 minutes.



In which two regions will the number of bacteria be highest?

- A S and T
- B S and U
- C T and V
- D U and V

Explanation: High light absorption in blue and red wavelengths → implies highest rate of photosynthesis → more bacteria

- 25 The copying of DNA is not always accurate, resulting in occasional, random changes to its base sequence. For example, scientists have calculated that one amino acid in vertebrate haemoglobin protein is changed every 10 million years.

Which statement about this process is correct?

- A A single amino acid change in a protein may be affected by natural selection.
- B All changes in the base sequence of DNA are affected by natural selection.
- C Natural selection can cause some of the amino acid changes in proteins.
- D The variability of DNA in a population will be less than the variability of its proteins.

Comments

A- One aa subject to NS to select for the haemoglobin protein

B – change in base sequence due to mutation, Silent mutation not affected by NS

C - change in aa due to mutation

D – no link to the process

- 26** In 1954, copper waste in the Finniss River killed numerous fish. This caused various species in the area to die out. However, one species, the black-banded rainbow fish, increased in numbers. The black-banded rainbow fish have modified gills that enable the fish to filter and remove the copper before it enters their body.

With respect to the black-banded rainbow fish it is reasonable to conclude that

- A** a mutation occurred in their population in 1954.
- B** the ability of their gills to remove copper already existed in 1954
- C** the high levels of copper in the water changed the structure of their modified gills
- D** their genomes are identical with those of the other species of fish that existed in 1954.

Comments : there existed the black banded rainbow fish that has the ability to remove copper within the population already. Mutation occurred prior to change in environment

- 27** Climate change has led to increasing temperature. The environments where birds, fishes, and other marine species live have become warmer. What is not an effect of climate change on these organisms?

- A** Certain fish species migrate in response to seasonal temperature changes, moving northward or to deeper, cooler waters in the summer and migrating back during the winter.
- B** The birds are moving north to feed and breed in the summer, then moving further south to spend the winter in warmer areas.
- C** The birds follow a regular seasonal migration pattern during wintering periods.
- D** As smaller marine prey species shift their habitats, larger predator species may follow them.

Comments

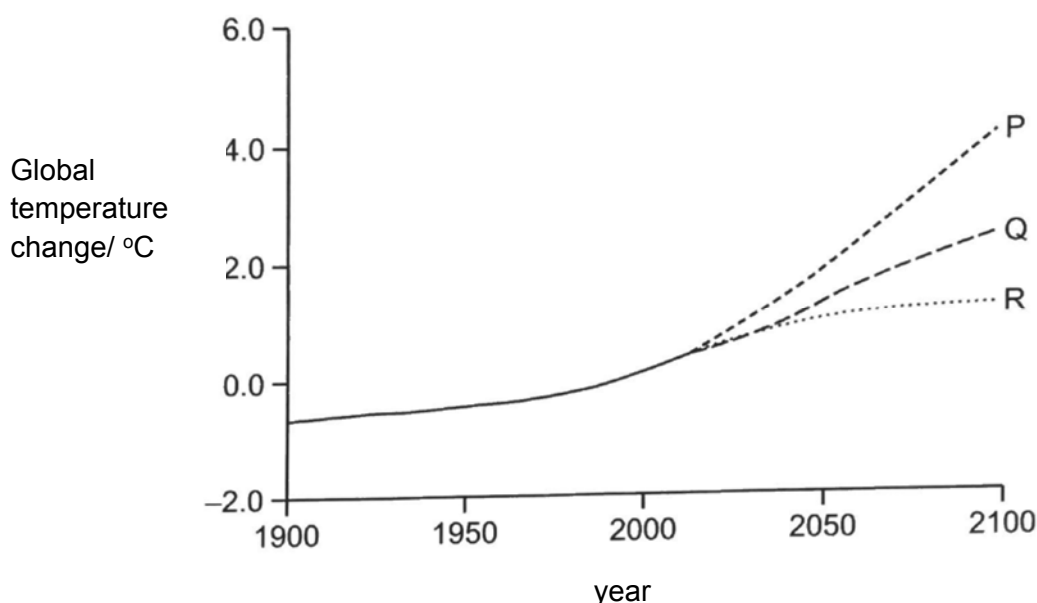
A, B and D are effect and consequences from an increase in temperature . Option C is the habit of birds and not an effect

- 28** Other than melting land-based ice sheets, which of these factors has made the largest contribution to the rise in sea level over the past 100 years?

- A** Warming of ocean surface water
- B** Melting sea ice
- C** Increased river runoff
- D** Increased reflection of infrared radiation

Explanation: sea ice is formed from sea water, therefore melting of sea ice will not rise sea level.

- 29 The graph shows the predicted changes in global temperatures using three different models, P, Q and R. Model Q assumes that no new factors act to influence the rate of climate change.



The predictions based on models P and R can be explained using some of the following statements.

- 1 An increased global temperature and reduced rainfall will lead to an increase in forest fires.
- 2 Permanently frozen soil and sediment in the Arctic will begin to thaw as global temperature increase.
- 3 Rising sea temperature will cause increase growth of photosynthetic algae.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support predictions P and R?

	statements that support prediction P	statements that support prediction R
A	<u>1, 2 and 4</u>	<u>3</u>
B	1 and 3	2 and 4
C	2	1, 3 and 4
D	3 and 4	1 and 2

Comments: Curve P shows a higher increase in global temperature. Statements 1, 2 and 4 lead to sharp increase in global temperature. Algal blooms absorb sunlight, making water even warmer and promoting more blooms.

30 Corals are among the first indicators of climate change. When ocean temperatures get too hot for too long, corals undergo a process called bleaching. Which statements are true about this process?

- 1 Increased levels of sediment in seawater, the zooxanthellae may lose substantial amounts of their photosynthetic pigmentation, which decreases rates of photosynthesis produces bleaching.
- 2 Bleaching occurs when abnormally high sea temperatures cause corals to expel the zooxanthellae living in them.
- 3 zooxanthellae unable to use the oxygen and waste materials of the host, supplying carbon dioxide and food substances in return.
- 4 With high sea temperatures and decreasing planktons, corals use the zooxanthellae as their food source

A 1 and 2

B 1 and 4

C 2 and 4

D 2 and 3

Statement 3 is wrong as zooxanthellae supply oxygen (from photosynthesis) to the corals and coral supply carbon dioxide to zooxanthellae.

Statement 4 is wrong as zooxanthellae is expelled out.